

Process Safety Fundamentals



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About

Designed to support those working in front-line operations, maintenance, and well related activities, the IOGP Process Safety Fundamentals are informed by data and designed to draw attention to situations that are most likely to lead to process safety event fatalities. They are not intended to exhaustively address all process safety risks and hazards in the oil and gas industry, but to be deployed in addition to a company's underlying systems for process safety management.

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Process Safety Fundamentals

Revision history

VERSION	DATE	AMENDMENTS
1.0	October 2020	First release
2.0	July 2026	Updated with additional data analysis and implementation guidance. No changes to individual PSF wording.

Contents

Introduction	5
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1. The Process Safety Fundamentals	8
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2. Process Safety Event Data Analysis	11
2.1. Background	11
2.2. Fatal Tier 1 PSE	11
2.3. Non-fatal Tier 1 PSE	13
2.4. Conclusion	14

3. Implementing the Process Safety Fundamentals	15
3.1. Where to apply the PSFs	15
3.2. How to implement and embed the PSFs	16
3.2.1. Sponsorship and Commitment	16
3.2.2. Developing	17
3.2.3. Implementing:	19
3.2.4. Embedding:	20
3.2.5. Monitoring and follow-up:	21
3.3. Potential additional PSFs	22
3.3.1. Operations	22
3.3.2. Wells	23

4. PSF Guidance Cards	24
------------------------------	-----------

5. Definitions	35
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Introduction

The IOGP Process Safety Fundamentals (PSFs) were developed in 2020 based on an analysis of data reported by IOGP Members between 2007 and 2017 which showed that 128 people had lost their lives in 56 process safety events during this period. In response to this, the IOGP PSFs were developed as a resource to support companies to reduce, and ultimately eliminate, fatal and high severity process safety events. As part of the update to this Report in 2026, the data analysis has been extended to include data up to and including 2024; within this period a further 30 PSE's resulted in 47 people losing their lives. See Section 2 for more information on this updated analysis.

Designed to support those working in front-line operations, maintenance, and well related activities, the IOGP PSFs are informed by data and designed to draw attention to situations that are most likely to lead to process safety event fatalities. They are not intended to exhaustively address all process safety risks and hazards in the oil and gas industry, but to be deployed in addition to a company's existing systems for process safety management.

IOGP PSFs are not intended to replace company policies, management systems, safe systems of work, safety training programmes, operating procedures, or work instructions; but to enhance them by reinforcing key aspects that have been identified as causal (in the process safety event fatalities reported to IOGP). In this respect, the PSFs rely on suitable management systems being in place.

Process safety is a disciplined framework for managing the hazards that have the potential to result in a major incident. Process safety is concerned with the integrity of operating systems and processes that handle hazardous substances and the hazards that arise from them.

The content of the PSFs should be familiar to the industry, but defining the fundamentals (as presented in this Report) provides a clearer, more direct, focus for those working in roles which have influence on the management of process safety risk on the most important aspects to be considered in their day-to-day activities. The roles which have an influence on process safety risk will include those who work in front-line operational or maintenance roles and technical personnel in roles which support the front-line (e.g., engineering support, technical authorities, project roles, etc.). Roles which influence process safety risk less directly (e.g., procurement, supply chain, planning, etc.) should also take notice of selected PSFs, as relevant to the scope of their involvement, and consider how these should be used to improve processes to manage risk.

The PSFs do not replace any of the IOGP Life Saving Rules (LSRs): they are intended as additional, complementary, recommendations. It is recommended that the PSFs are kept separate from the LSRs and not combined into a single list. The reason for this is that the LSRs are primarily focused on personal safety events, with some overlapping into process safety, whereas the PSFs are specifically targeted at the causes of process safety events. Although there is overlap between some LSRs and PSFs (e.g., Energy Isolation/We Maintain Safe Isolation and Hot Work/We Control Ignition Sources), it was considered necessary to retain additional complementary information within the PSFs as these are identified causes of process safety events. Member company experience demonstrates that the PSFs and the LSRs can be used effectively in a complementary way.

Table 1 - Life-Saving Rules and Process Safety Fundamentals.

Life-Saving Rules		Process Safety Fundamentals
Focus	Strong focus on personal safety with some overlap into process safety	Solely focused on process safety
Flexibility	Single common industry set recommended by IOGP	Core set recommended by IOGP, with flexibility to supplement or exchange a small number based on local process safety challenges

The PSFs provide focus on supporting the front-line to enable successful process safety performance (see IOGP Report 456 – *Process safety - recommended practice on key performance indicators*). If activities cannot be carried out in line with these PSF expectations and behaviours, employees are expected and encouraged to speak up. Any difficulties in applying the PSFs should be openly voiced and reported without fear of criticism or negative consequences, so that obstacles to implementation can be understood and addressed. Bringing these issues into the open enables the company to decide on how to address the concern raised. It is important that front-line workers are given the flexibility to use the PSFs as a supportive tool to help improve process safety performance. The PSFs are not intended to be seen as ‘another set of rules’, and it is strongly advised that they are not associated with disciplinary measures for non-conformance as this approach to launching the PSFs may negate potential benefits that could be gained by the programme.

IOGP is seeking industry standardization by recommending a core set of 10 PSFs relevant to all upstream operations. It is recognized, however, that some companies may wish to exchange a small number of the core set for others that are more tailored to their company’s specific operations and process safety challenges or to create standardization on PSF across different businesses (e.g., downstream) within an integrated company. In doing so, companies are encouraged to align with the core set as far as possible and exercise caution in removing core PSFs given they have been selected based on thorough data analysis. A possible list of additional PSFs, based on existing Member Company deployments and feedback, is provided in Section 3.

Fatal wells-related (i.e., drilling, work-over, etc.) process safety events were included in the original analysis to develop the IOGP PSFs, so it is believed the PSFs are also relevant to that area of operation. Companies who wish to apply the PSFs to well operations may consider including supplementary PSFs to address wells-related operational activities.

It is intended that adoption of PSFs:

- Provides a focus for process safety at the front-line.
- Enables process safety concerns to be raised and addressed.
- Is a step towards industry-wide common language for process safety.
- Helps companies to eliminate high severity process safety events.

This document provides the Process Safety Fundamentals, the supporting data analysis, and implementation guidance.

IOGP welcomes feedback on the Process Safety Fundamentals (PSFs) and will continue to review process safety event data. These two sources of insight will guide any future decisions on whether changes to the core set of PSFs are needed. At the time of this Report update, no changes to the core PSFs have been identified as necessary.

At future updates of the Report, IOGP may update the core set of PSFs based on real-world application feedback from IOGP Member companies, including any of the following:

- Effectiveness of the 10 core PSFs.
- Other PSFs that have proved useful to IOGP Member companies.
- Adaptations or supplementary PSFs that have been helpful to include downstream and wells activities.

Additional PSF implementation resources are available at www.iogp.org/psf.

1. The Process Safety Fundamentals



We respect hazards

- We improve our understanding of process safety hazards at our location and our roles in controlling them.
- We are vigilant about the potential impacts of uncontrolled process safety hazards.
- We discuss process safety hazards before starting a task.
- We bring forward process safety hazards to be included in activity risk assessments.



We apply procedures

- We use operating and maintenance procedures, even if we are familiar with the task.
- We discuss the key steps within a critical procedure before starting it.
- We pause before key steps and check readiness to progress.
- We stop, inform supervision and avoid workarounds if procedures are missing, unclear, unsafe, or cannot be followed.
- We take time to become familiar with, and practice, emergency procedures.



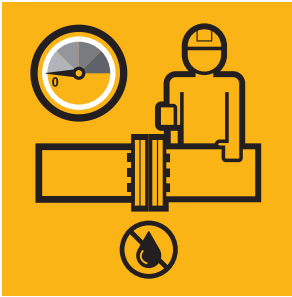
We sustain barriers

- We discuss the purpose of hardware and human barriers at our location.
- We evaluate how our tasks could impact process safety barriers.
- We speak up when barriers don't feel adequate.
- We perform our roles in maintaining barrier health and alert supervision to our concerns.
- We use an approval process for operations with degraded barriers.



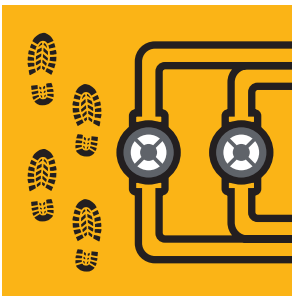
We stay within operating limits

- We discuss and use the approved operating limits for our location.
- We escalate where we cannot work within operating limits.
- We alert supervision if an alarm response action is unclear or the time to respond is inadequate.
- We obtain formal approval before changing operating limits.
- We confirm that potential for overpressure from temporary pressure sources has been addressed.



We maintain safe isolation

- We use isolation plans for the specific task, based on up-to-date information.
- We raise isolation concerns before the task starts and challenge when isolation plans cannot be executed.
- We check for residual pressure or process material before breaking containment.
- We monitor the integrity of isolations regularly and stop to reassess when change could affect an isolation integrity.
- We confirm leak-tightness before, during, and after reinstating equipment.



We walk the line

- We use up-to-date documentation (e.g., P&IDs)¹ that accurately reflect installed systems and equipment.
- We physically confirm the system is ready for the intended activity (e.g., valve positions, line up of relief devices, etc.).
- We alert supervision to identified documentation and readiness issues before operation.



We control ignition sources

- We identify, eliminate, or control the full range of potential ignition sources during task risk assessments and during job preparation and execution.
- We minimise and challenge ignition sources even in “non-hazardous” areas.
- We eliminate ignition sources during breaking containment and start-up and shutdown operations.



We recognise change

- We look for and speak up about change.
- We discuss changes and involve others to identify the need for management of change (MOC).
- We review the MOC process for guidance on what triggers an MOC.
- We discuss and seek advice on change that occurs gradually over time.

¹ Piping and Instrumentation Diagrams



We stop if the unexpected occurs

- We discuss the work plan and what signals would tell us it is proceeding as expected.
- We pause and ask questions when signals and conditions are not as expected.
- We stop and alert supervision if the activity is not proceeding as expected.



We watch for weak signals

- We proactively look for indicators or signals that suggest future problems.
- We speak up about potential issues even if we are not sure they are important.
- We persistently explore the causes of changing indicators or unusual situations.

Guidance cards

A set of guidance cards to support implementation of the PSFs is available in Section 5. These have more detailed descriptions of the activities, definitions, and reference materials, and can support the understanding and implementation of the PSFs. Companies can use these to provide additional help and guidance to the front-line workforce, but they are more specifically intended for the supporting organization, particularly supervisors and managers.

2. Process Safety Event Data Analysis

2.1. Background

The core set of PSFs were originally developed based on Process Safety Event (PSE) data reported by IOGP Member Companies over a period of eleven years (2007-2017). This data showed that 128 people lost their lives in 56 Process Safety Events.

Analysis of the process safety related fatal incidents was conducted by a team of subject matter experts formed from the IOGP Process Safety Subcommittee (PSSC) with the aim of determining what types of Process Safety Fundamentals (PSF) were a factor in the incidents and how, if they had been implemented effectively, they might have prevented or mitigated the fatal consequences of these incidents.

The analysis considered a broad range of potential PSFs, influenced by both the information on the fatal incidents provided by IOGP Member Companies, and the team members' experience. Each fatal PSE was reviewed and linked to one or more potential PSF. By determining which PSF could be linked to either the highest number of fatal incidents or number of fatalities allowed a shortlist of PSF to be identified which eventually became the core set of 10 IOGP PSFs.

This Report extends the analysis to include data reported by IOGP Member Companies for the years 2018–2024. All reported data, including cases where PSFs were already assigned to incidents (by the reporting Companies) in recent years has been reviewed and validated by subject matter experts from the IOGP Process Safety Sub-Committee (PSSC). The results of this updated analysis are presented in the following Sections.

2.2. Fatal Tier 1 PSE

To update this analysis, data for PSE involving fatalities during the years 2018-2024 were added to the original dataset². The extended dataset shows that 175 people lost their lives in 86 PSEs over the 18-year period 2007-2024.

IOGP Member Companies provided sufficient detail in 72 of the 86 incident reports to allow a reasonable assessment linking each event to one or more specific PSFs. As shown in Figure 1, 94% of these fatal PSEs (68 out of 72 incidents) could be associated with one or more of the core set of 10 IOGP PSFs.

² Participation in IOGP's annual collection of safety and process safety performance data is voluntary, and not all member companies contribute to every dataset. In addition to the PSE data used in this analysis, three additional Tier 1 PSEs with fatal outcomes were reported in the safety performance dataset (IOGP Reports 2024s and 2024sf) but not in the process safety dataset, as the reporting companies did not participate in the specific PSE data collection.



Figure 1 - Proportion of fatal PSEs linked to one of the PSF, 2007-2024.

Figure 2 illustrates how the 68 fatal PSEs were linked to one primary PSF. Although some incidents were associated with multiple PSFs, Figures 2 and 3 show only the PSF that was considered the primary contributor to the incident for clarity.

Two of the ten PSFs – ‘Respect hazards’ and ‘Report weak signals’ – were not part of the original data analysis but were added later in the selection process when it became clear they were underlying causes in many incidents.

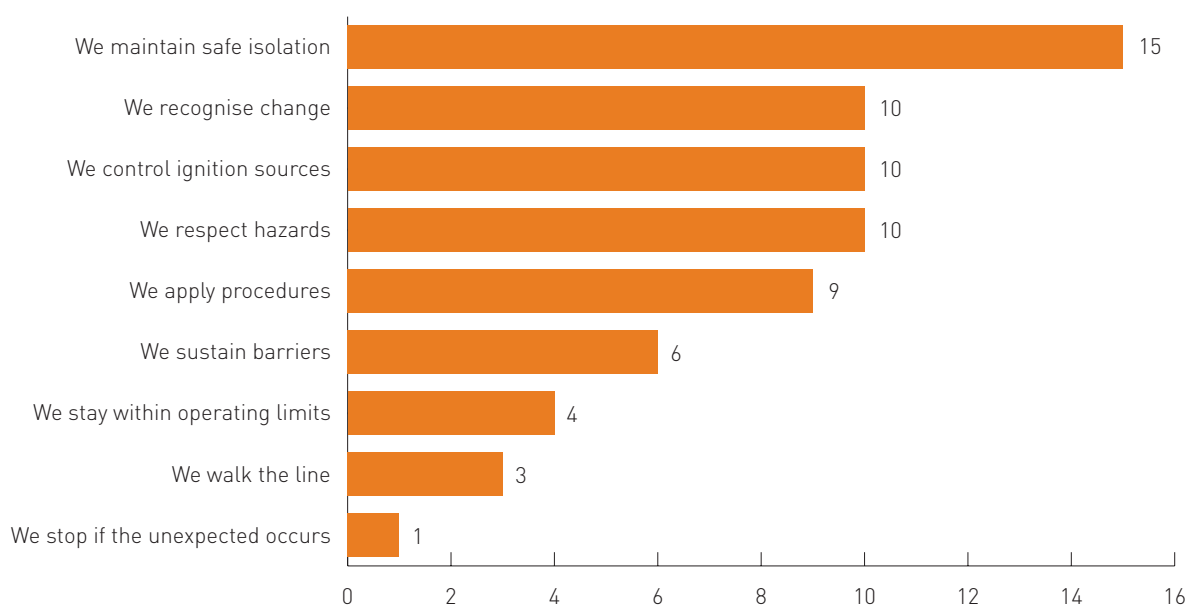


Figure 2 - Number of fatal PSE and the related primary PSF, 2007-2024.

A total of 175 workforce fatalities were associated with the 86 incidents included in the analysis. Of these, 116 fatalities resulted from the 68 PSEs reported with sufficient detail to be linked to one or more PSFs. Figure 3 shows how these fatalities were distributed across the core set of 10 PSFs.

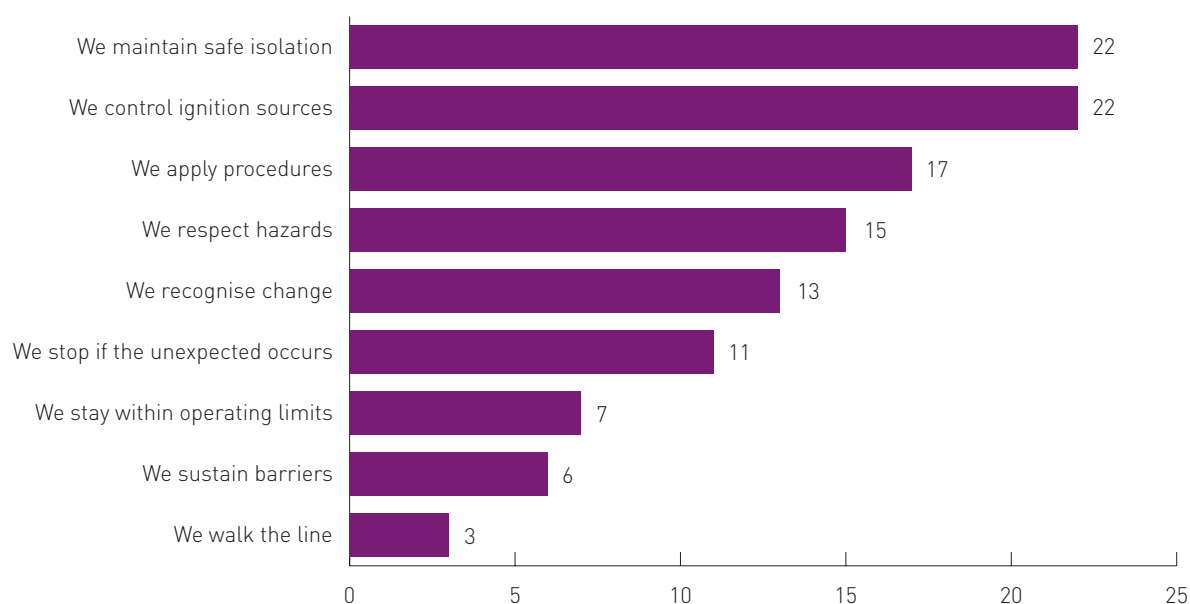


Figure 3 - Number of fatalities and the related primary PSF, 2007-2024.

Figures 2 and 3 show that the primary PSFs linked to both the largest number of fatal PSE incidents and the highest number of fatalities are:

- We maintain safe isolation
- We control ignition sources
- We apply procedures
- We respect hazards
- We recognise change

2.3. Non-fatal Tier 1 PSE

IOGP members were first requested to include information on the applicable Process Safety Fundamental (PSF) as part of the Tier 1 PSE data submission in 2021. Between 2021 and 2024, Member Companies reported 781 Tier 1 PSEs. Ten of these incidents involved one or more fatalities, leaving 771 non-fatal Tier 1 PSEs.

For 61 of these non-fatal events, no PSF could be specified due to insufficient information provided by the reporting company. For 215 (30%) there was no applicable PSF because these events were primarily caused by design issues or mechanical failures (e.g., corrosion and erosion).

The other 495 (70%) events could be linked to one or more PSFs, as shown in Figure 4. The percentage of 70% is high, as although the PSFs are primarily aimed at eliminating fatal PSEs, they also influence a significant proportion of non-fatal PSEs - as would be expected. This percentage is lower than for fatal PSEs during the same period (93% for 2021–2024), mainly because the non-fatal dataset includes a higher proportion of mechanical failure events to which PSFs do not normally apply.

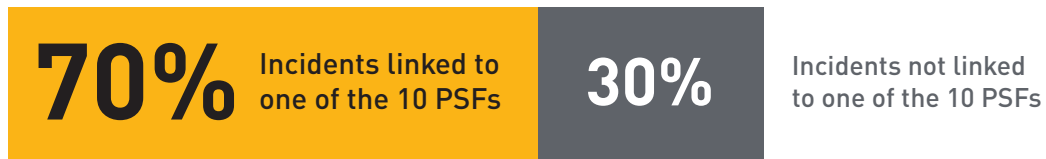


Figure 4 - Proportion of non-fatal PSEs linked to one of the PSF, 2021-2024.

2.4. Conclusion

As a conclusion of the analysis of the extended dataset (2007–2024), there is no strong indication that changes to the core set of 10 Process Safety Fundamentals (PSFs) are necessary – either through the addition of new PSFs or revisions to the existing wording. Therefore, the original core set of 10 PSFs continues to represent the recommended practice for supporting companies in reducing – and ultimately eliminating – fatal and high-severity process safety events.

3. Implementing the Process Safety Fundamentals

3.1. Where to apply the PSFs

The PSFs are not intended to replace company safe systems of work, operating procedures, work instructions, etc. – but to enhance them by reinforcing key aspects. The PSF are intended for front-line personnel but can also be applied by supporting functions, e.g., in work planning activities. The following are examples of the way in which the PSF can be implemented in typical activities.



Work Planning / Pre-job risk assessment

- Are we doing any work involving a PSF?
- What process safety barriers could be affected by the work we are planning?
- What do we need to do to follow the PSF from start to finish?
- What controls will need to be in place?
- What needs to be communicated? Who to?



Pre-job Toolbox talk/safety meeting

- Is everything as described in the job plan?
- Does everyone in the work team understand the job, the risk controls, and PSFs to be applied?
- Is everything in place, and in good working condition?
- What checks, trials, tests, drill exercises need to be done?



Last-minute risk assessment

- Have we completed all the PSF actions?
- Are there any hazards or ignition sources we didn't identify?
- Re-assess the risk pre-task and during the job (e.g., between task steps) or if anything changes.
- Are we monitoring for weak signals which may indicate a problem?



Post job review

- Did we take all the actions associated with the PSFs?
- What went well? What didn't go well?
- Anything to note for the next time we perform this task or work in this area?
- Update procedures, risk assessments, etc. Don't leave it to memory.



Observations and walkabouts

- Check the PSFs during walkabouts and field observations
- Do you see anyone performing work where a PSF is relevant?
- Are they following the PSF?
 - Yes? Great, recognize it!
 - No? Intervene!



Intervention

- Intervene or STOP WORK if a PSF is not being followed or could be done better.
- DO NOT START WORK if the requirements cannot be met – ask for support.



Incident Investigation

- What can we learn from incidents that involved a PSF not being followed or where a PSF prevented the incident from being worse? How can we share these learnings to improve PSF implementation?

3.2. How to implement and embed the PSFs

A key success factor in the implementation and embedding of the PSFs is not just what is rolled out, but how it's done – the manner and sponsorship are critical. To support a comprehensive and effective PSF programme at site or plant level, the following steps (see Figure 5) are proposed:

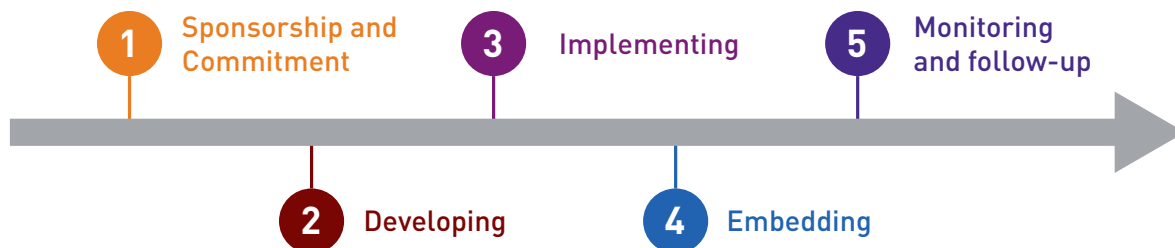


Figure 5 - PSF Implementation 'Roadmap' of key steps.

3.2.1. Sponsorship and Commitment

Suggested areas of focus:

- Engage and align leadership around the value of adopting the PSFs.
- Identify and confirm a senior executive or leader to sponsor the PSF programme.

Sharing PSF materials at the site or plant level, with involvement from relevant groups and sponsors, serves as an effective starting point. A half-day workshop can be organized, gathering key representatives from HSE and process safety, asset integrity, production, operations, maintenance, and well operations. Senior management participation is essential for successful programme implementation, as it underscores that process safety is a collective responsibility and not solely the remit of frontline personnel.

The workshop should focus on two primary objectives. First, present the PSFs to provide an overview of their background and the benefits of adoption. Second, adopt an interactive approach to solicit feedback from the front-line regarding effective implementation of the PSFs within their respective sites or plants.

To encourage open discussion, consider using questions such as:

- Why are PSFs important to you and the organization?
- How can PSFs be effectively implemented at your location?
- What changes are needed to drive process safety improvement?
- In what ways can leadership further support front-line teams in applying the PSFs?

One key outcome of the engagement workshop is to formally agree on a proposed plan for PSF implementation, based on the key steps outlined in this Section. Feedback from organizations that have already adopted the PSFs indicates that the designation of a senior leader as programme sponsor is a critical decision point. This demonstrates leadership commitment and provides vital support in reinforcing the PSFs during rollout.

Case Study Examples:

Example #1:

Company A launched a comprehensive internal workshop for leadership to discuss the integration of PSFs into daily operations. The workshop featured expert speakers who presented case studies from major incidents, demonstrating the importance of a robust process safety culture. Following the session, company A's leadership team committed to launching a global process safety initiative, including frequent reviews of safety protocols and regular engagement with front-line staff.

Example #2:

Company D leveraged a refresh of its corporate safety strategy to introduce the PSFs, a strategy championed by the CEO and embraced across all levels of leadership.

3.2.2. Developing

Suggested areas of focus:

- The development of a sponsor-endorsed PSF plan, including scope, implementation, embedding, monitoring, and follow-up activities.
- The creation of supporting materials for PSFs, such as rollout content, posters, booklets, videos, pocket cards, and related resources.

Based on the local context, companies are strongly encouraged to remain consistent with the 10 core PSFs but have the flexibility to integrate additional PSFs (or exchange a small number of the core 10) based on specific local process safety challenges. Any decision to remove core PSFs should be approached with caution, as these elements have been identified through rigorous analysis of data. Adoption of the core PSFs contributes significantly to industry standardization and the establishment of a common language around process safety. In cases where companies elect to substitute one or more core PSFs, such decisions should be supported with a thorough analysis of its internal incident and leading indicator data related to actual or potential high consequence process safety events.

The duration required to complete the developing phase may vary depending on factors such as organizational size (e.g., single site versus multiple sites; small, simple assets versus large, complex ones), types of agreements in place (such as production sharing agreements or joint ventures), and the number of parties involved. Establishing a sponsor-endorsed PSF plan serves as an effective means to document key decisions, strategic objectives, and supporting actions, thereby aligning stakeholders and ensuring early leadership engagement. This framework promotes ownership and provides clarity throughout the implementation process.

For effective implementation, it is advisable to appoint one or more PSF promoters or champions to oversee the rollout, with their responsibilities clearly outlined in the implementation plan. During the development phase, these individuals can provide support by collecting feedback from front-line workers regarding the practical application of PSFs at each location. Data analysis the selection of the PSFs is based on can be shared, and opinions and suggestions from the front-line can be collected.

Examples of supporting material content:

- **Videos:** Informative videos can help increase attention and information retention. They may improve audience understanding, prompt questions, and encourage discussion. Short videos could be produced for each PSF to describe its significance and outline the roles employees play in preventing process safety incidents. Companies should seek to use social media and other communication methods to share information.
- **Posters:** Posters serve as effective communication tools by presenting relevant images at key locations. Selected visuals can communicate ideas clearly alongside written information.
- **Booklets/brochures:** A compact booklet on PSFs is a useful resource for maintaining awareness and stimulating process safety discussions, such as during toolbox talks or operational site visits conducted by managers and supervisors.

Case Study Examples:

Example #1:

Company B adopted all the PSFs, retaining the original texts but including some additional text to address company standards and local process safety challenges. All PSFs have been translated into local languages for adoption at company B's operating sites.

Example #2:

Company C performed a gap analysis of its process safety management systems following several high-profile incidents in the oil and gas industry. The gap analysis led to a comprehensive overhaul of their safety protocols, leading to the adoption and promotion of PSFs. Initial awareness sessions were delivered to all teams to support the rollout, followed by a sustained effort to reinforce consistent messaging. This included regular communication through the monthly internal newsletter, which highlighted examples of positive PSF behaviours demonstrated by front-line employees and linked these to key lessons from industry-wide process safety incidents.

Example #3:

Company D initiated the "pilot phase" of its PSFs implementation at three selected sites. Feedback was collected through questionnaires from these sites, providing valuable insights that helped refine the implementation approach. Building on the pilot's success, the PSF dissemination campaign was subsequently rolled out across all company D sites the following year.

3.2.3. Implementing:

Suggested areas of focus:

- Initiation of PSF at the plant or site, with sponsors affirming commitment to its adoption.
- Commencement of PSF material rollout across the plant or site.

The process commences with an official presentation of the PSF programme and its components to plant or site leadership. Engaging the sponsor to introduce the initiative and underscore their commitment is essential for achieving alignment and facilitating a successful launch.

The rollout phase should be regarded as an ongoing engagement activity across the plant/site. Each plant/site may select the most appropriate tools and methodologies, such as:

- Targeted coaching sessions on PSFs, facilitated by promoters or champions. These sessions aim to enhance front-line workers' understanding of the programme's scope and objectives, while also providing opportunities for additional guidance and clarification.
- Awareness campaigns: regular distribution of PSFs to front-line staff, including references to actual events at the site and incorporating lessons learned as well as relevant industry materials.

Case Study Examples:

Example #1:

Company A's EVP officially launched the PSF programme and delegated PSF implementation directly to site managers (instead of site-based process safety managers), positioning site leadership as directly accountable for embedding the fundamentals into day-to-day operations.

Example #2:

Company B's CEO introduced the PSFs during their global Safety day. This was followed by a series of (approximately monthly) small group discussion sessions led by local managers/supervisors across all operated sites in which presentations, linking the PSFs to historical incidents within the company, were used to stimulate conversations. Videos and presentation materials were produced by the BU teams under the coordination of the corporate safety team. The group discussions and the use of historical company incidents made the message both relevant and personal, highlighting how the application of PSFs might have altered the outcomes of those events.

Example #3:

Company C introduced a 'safety ambassador' programme, where employees at all levels were trained to be champions of process safety within their departments.

Example #4:

Company D initiated a “pilot phase” at three selected sites. At each site, designated PSF promoters led an on-site dissemination campaign, aimed at reaching all operational personnel.

Example #5:

Company F enhanced its process safety communication by incorporating PSFs into their process safety lessons posters. Each lesson learned shared is linked to a relevant PSF, creating awareness and understanding of PSFs among personnel.

3.2.4. Embedding:

Suggested areas of focus:

- The PSFs can be incorporated into day-to-day work processes (including procedures, checklists, templates, and tools) to maintain consistency and sustainability.
- PSF are included in new hire inductions, training and contractor onboarding.

Examples of embedding activities can include:

- Integration into existing systems and processes: Embedding PSFs into current activities and communications, such as shift routines, maintenance planning, and permit to work procedures, may improve implementation efficiency and sustainability.
- Recognition and interventions: Site hazard observation cards list the PSFs, and company programmes encourage the use of PSFs through recognition and awards for employees. PSFs may also be integrated into pre-job/task risk assessment templates and toolbox talks materials to ensure discussion and understanding of the necessary controls prior to work commencement.
- Site visits and inspection programme: Supervisors and managers are expected to conduct site visits to monitor PSF implementation. Issues raised by front-line workers can be reviewed to support ongoing improvements in process safety performance.
- Integration into contractual agreements: PSFs adopted by the Company may be referenced within bridging documents, either directly or indirectly, ensuring that Contractors working on Company sites are informed of PSF inclusion in Company policies and procedures. Integration into contractual agreements: Referencing PSFs within bridging documents ensures contractors understand and apply them. Including PSFs in these documents or related contractual agreements clarifies expectations and supports consistent integration of PSFs in how contractors plan and execute their work while under contract with the company.
- Plant/site inductions: Incorporating the PSF into the induction process ensures that all individuals attending the site receive an introduction to the PSF, as well as an understanding that the PSF has been adopted and implemented at the location.
- Reviews and audits: incorporating the PSFs into check-sheets and audit protocols will help to ensure that they are discussed during reviews and audits.

Case Study Examples:

Example #1:

Company B issued the 'HSE expectation booklet' to all personnel across all assets, focusing on the company's HSE values and incorporating the PSFs as well as Life Saving Rules. The booklet outlines the company's expectations for pre-job planning, risk assessment, toolbox talks, stop work, etc., providing references for personnel to discuss the safety expectations of each task with their supervisors. The HSE expectation booklet is translated into local languages to enable company-wide implementation and adoption.

Example #2:

Company F integrated PSFs into its existing systems and processes by embedding PSFs within the Job Safety Analysis (JSA) form. This enables personnel to identify and discuss relevant PSFs in relation to the specific task being assessed.

Example #3:

Company G has integrated PSFs into their site safety observation campaign, which promotes proactive behaviours by encouraging personnel to speak up whenever they see unsafe practices or feel unsafe in their workplace. This inclusion enables personnel to link safety observations with related PSFs and engage in constructive dialogue at the workplace.

3.2.5. Monitoring and follow-up:

Suggested areas of focus:

- Analysis of PSF-related performance data, including incidents, losses of primary containment, near misses, and other leading indicators.
- Verification of implementation in line with the agreed plan.

Measuring the outcomes of implementing PSFs within an organization may require some time. Since fatal Process Safety Events (PSEs) are infrequent within individual companies, it is often more instructive to assess improvements through data related to loss of primary containment, process safety near misses, and other suitable leading indicators related to barrier health.

When evaluating process safety trends internally, it is useful to categorize different event types and to exclude those related to design integrity or technical integrity; PSFs specifically target events linked to operating integrity. Although statistically significant reductions in PSEs may take years to materialize, a noticeable increase in front-line awareness, competence, and engagement in process safety can typically be observed much sooner.

In the short term, organizations may consider tracking near-miss events and site improvement suggestions through to resolution. This approach not only supports workforce engagement with the PSFs but also enables effective monitoring of programme health. Furthermore, gathering data on instances where front-line personnel are unable to fulfill

PSF requirements – such as constraints imposed by facility design or lack of comparable processes – can help organizations pinpoint process and design variances. Addressing these through engineering modifications will support more robust PSF implementation and overall process safety improvements. Encouraging the reporting of situations where PSFs are challenging to achieve, along with the reasons, further engages staff in daily operations, such as during toolbox talks and permit-to-work meetings.

Additionally, introducing a verification stage later in the programme provides a valuable opportunity to evaluate the effectiveness of PSF implementation against initial objectives. This process strengthens accountability, recognizes accomplishments, and identifies areas for ongoing enhancement and continuous improvement, especially for those companies at a mature stage of implementation.

Case Study Examples:

Example #1:

Company H established an internal system for tracking process safety metrics, including near misses and incident investigations. Regular audits were conducted, and a feedback loop, where field employees could report safety concerns directly to senior leadership, was established to ensure corrective actions were promptly implemented.

Company H reported a significant improvement in both near-miss reporting and incident reduction, attributing this success to continuous monitoring and the active involvement of all levels of the organization.

Example #2:

Company B has defined a process for reporting PSF deviations in relation to incidents and near-misses, and for barrier 'weaknesses' for hardware and documentation/procedure related barriers. Analysis of data collected from reporting provides valuable indicators of the effectiveness of specific PSFs.

3.3. Potential additional PSFs

Additional PSFs have been established by IOGP Members, examples of which are listed below. Many of the topics covered by these other PSFs are covered (directly or indirectly) by the IOGP recommended PSFs; however, companies may choose to adopt or adapt versions of these to complement the core set of PSFs:

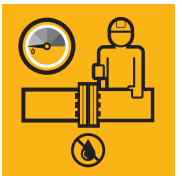
3.3.1. Operations

- We ensure continuous attendance at manual open draining, bunkering, and critical transfer operations.
- We control utility systems connected to a process.
- We manage risks associated with temporary equipment.
- We perform thorough and detailed crew/shift handover including clear communication of plant status and risks.

- We check the absence of an explosive atmosphere in furnaces before igniting burners.
- We avoid splash loading.
- We move non-essential workers away from the potential effects of hazards.
- We ensure that we always have sufficient competent personnel to safely operate and maintain our facilities and to respond to emergencies.
- We understand the scenarios where operator action or response to an alarm is considered part of a barrier and assure the competence and ability of operators to respond as required.
- We report incidents and near misses - investigate, share, and learn.
- We verify aviation fuel quality because contamination can lead to unsafe operating conditions, equipment damage, or loss of primary containment.

3.3.2. Wells

- We check well control equipment and confirm it is certified and tested.
- We ensure effective well isolation, with at least two barriers, when working downstream of a well.
- We “Walk the line” on temporary rig-ups and confirm setup against the layout drawing.
- We use at least two independent barriers for pressure containment and material process flow.



4. PSF Guidance Cards

WE RESPECT HAZARDS

- We improve our understanding of process safety hazards at our location and our roles in controlling them.
- We are vigilant about the potential impacts of uncontrolled process safety hazards.
- We discuss process safety hazards before starting a task.
- We bring forward process safety hazards to be included in activity risk assessments.



Incident investigations show that lack of hazard awareness is an underlying cause of many process safety events. It is important that we understand the process safety hazards that we face at our facilities and are aware of the harm they could generate if we lose control of them.

Typical hazards include flammable liquids and vapours, combustibles, toxic chemicals, asphyxiants, corrosives, pyrophorics, and high pressure/temperature.

When we routinely work close to process safety hazards it is possible to become desensitized to them; this can be called risk normalization. This can lead to us not treating the hazard with sufficient respect and us being less thorough about how we control them. Often, people underestimate the potential impact of process safety hazards, particularly if they have not been directly involved with a major accident. Pictures or videos of previous process safety incidents can help to reinforce the sheer size of the potential consequences of our hazards. It is good practice for teams to discuss the process safety hazards at their facility and whether they fully understand the potential impacts. It can be helpful to ask your company's process safety specialists to explain the facility's process safety studies.

Another common concern is that it can be easy to focus on personal safety issues at the expense of process safety. Sometimes personal safety issues are more obvious and readily spotted and that can lead to process safety hazards being missed. It is important to recognize that both personal and process safety are vitally important and it is recommended to give due attention to both. Teams are encouraged to be alert for risk assessments that do not cover all the process safety risks associated with an activity.

Depending on your company management systems and legislative framework you will find guidance on major hazard analysis in documents such as: Safety cases, PHAs (Process Hazard Analysis), safety reports, corporate risk summaries, and other internal resources.

Tips for Leaders and Managers:

Create space to discuss major accident hazards with front-line workers.

Review risk assessments to ensure they address both personal and process safety hazards – not just one or the other.

Encourage teams to stay alert to process safety hazards and avoid becoming desensitized to their work environment.

Attend toolbox talks to reinforce the importance of discussing process safety aspects of a task.

Discuss interrelated hazards and cumulative risks, especially where multiple barriers are interdependent.

Challenge high-risk work near or on live systems.

In the field, ask the frontline:

- How do you identify relevant process safety hazards in your risk assessments?
- What controls do you have in place to manage the process safety hazards associated with your task?

Additional guidance:

- Energy Institute: [Reflective Learning video – 'Removing the hazards'](#)
- Step Change in Safety: [Major Accident Hazard Understanding](#)
- Safer Together: [Process Safety Awareness](#)
- CCPS Beacon [Hidden Hazards! – August 2019 | English](#)
- US Chemical Safety Board: [Refinery Disaster in Philadelphia](#)
- US Chemical Safety Board: [Winterization Safety Message](#)

WE APPLY PROCEDURES

- We use operating and maintenance procedures, even if we are familiar with the task.
- We discuss the key steps within a critical procedure before starting it.
- We pause before key steps and check readiness to progress.
- We stop, inform supervision, and avoid workarounds if procedures are missing, unclear, unsafe, or cannot be followed.
- We take time to become familiar with and practice emergency procedures.



Many of the operations or activities we perform on our facilities are complex and/or have the potential to release hazardous materials if they are not performed correctly. Step by step procedures are developed to perform these tasks safely and prevent unwanted or hazardous situations. It is good practice to use procedures on site and use job aids (e.g., sign off) to confirm that key steps have been completed in the correct sequence as the activity proceeds. Each company or asset may want to consider which of its activities or tasks are “critical” and require the higher level of assurance that “sign off” for each step of a procedure gives to the process. Typically, activities such as startup/shutdown of a facility, or particularly hazardous activities like pigging, are considered “critical” procedures.

Before starting a procedure, it is good practice to discuss the task ahead and how it is expected to proceed. The team involved can discuss the key steps, particularly those which are irreversible, and what will be expected at those stages to confirm readiness to proceed further.

In most cases existing procedures will be accurate, however if you identify issues with the quality or completeness of the procedures such as a lack of clarity on a task or missing/incorrect data, then this should be raised so they can be addressed as part of your MOC process.

It is easy to become complacent about an activity that has been performed many times before without hazardous or unwanted outcomes. However, no matter how experienced we are, it is easy to make a mistake, and therefore it is important to apply the procedures thoroughly, every time.

If you cannot complete or follow the procedure as it is currently written or you think there is an issue with the procedure, halt the activity and raise the issue with your supervisor. If you think there is a better way of performing an activity/task discuss it with your supervisor and raise a formal change request.

Sometimes an activity or operation is not completed in one shift, and it therefore is important to ensure there is an effective shift handover process so that the new shift has accurate knowledge of the status of operations and any issues they should be aware of.

If a hazardous situation occurs, it is also important to understand and apply emergency response procedures. These need to be readily usable in more stressful situations and regular practice drills help to reinforce understanding and familiarity.

Tips for Leaders and Managers:

Verify procedures are up-to-date, accurate, and effective, and they reflect the actual task or activity being performed.

Procedures should be easy to use and readily available to those who need them.

Confirm human factors are adequately considered in procedure design.

Plan that personnel have sufficient time to become familiar with the plant, its equipment, and its procedures.

Ensure that any issues raised are addressed and that procedures incorporate all relevant learnings.

In the field, ask the frontline:

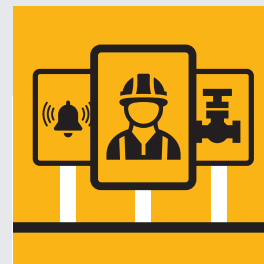
- Are your procedures adequate and easily available?
(Provide feedback to the procedure owner or responsible discipline.)
- Are there any workarounds used? Why are they needed? What is the underlying issue?
(Discuss how procedures can be updated to reflect actual work practices.)
- Are you familiar with the sites Emergency Response procedures? What is your role?

Additional guidance:

- Refer to local company guidance on procedures and consider linking to Human Factors guidance
- Center for Chemical Process Safety (CCPS). *Guidelines for Writing Effective Operating and Maintenance Procedures*. New York. 1996.
- US Chemical Safety Board: [Baton Rouge Refinery Chemical Release and Fire | CSB](#)

WE SUSTAIN BARRIERS

- We discuss the purpose of hardware and human barriers at our location.
- We evaluate how our tasks could impact process safety barriers.
- We speak up when barriers don't feel adequate.
- We perform our roles in maintaining barrier health and alert supervision to our concerns.
- We use an approval process for operations with degraded barriers.



Most companies find it helpful to think about the mental model of the barriers that we use to control process safety hazards. A barrier is a risk control that prevents unintended events from occurring or stops the escalation of harmful consequences.

We can think about barriers as being either hardware or human barriers. Hardware barriers include primary containment, process equipment, and engineered systems designed and managed to prevent hazardous releases or mitigate their consequences. Although people are involved with maintaining and testing hardware barriers, they do not rely on someone taking action when a demand is placed on the barrier. Human barriers, however, do rely on the actions of people when a demand is placed on the barrier, for example response to a process alarm. Both types of barriers are important, but hardware barriers are typically considered to have greater reliability, if maintained correctly.

It is good for front-line teams to understand and discuss the barriers that are in place on their facility, and how reliable and effective they are. It is important that individuals feel able to speak up if they believe that a barrier is weak or inadequate. This can be also done by practicing independent verification by co-workers.

Many find that a bowtie diagram is a helpful visual aid when having this conversation. Another helpful tool is to develop and provide a simplified scenario diagram showing the equipment and important barriers contained within the scenario.

Understanding the role that barriers play, and what could potentially happen if a barrier fails, is important as it allows teams and individuals to understand the influence they have in sustaining barrier health.

Process safety events can result from degraded or failed barriers. Degraded barriers include those that are inhibited, overridden, bypassed, unreliable, or unavailable. These should be addressed without delay and normally require approval for continued operations. Often front-line workers will play a key role in sustaining and monitoring any additional mitigations that are put in place whilst a degraded barrier is being restored.

Tips for Leaders and Managers:

Provide tools (e.g., bowties) to enable front-line workers to understand and visualize process safety barriers at their location.

Ensure it is known who is accountable/responsible for each barrier's performance.

Implement systematic barrier management processes and allocate sufficient resources to test and maintain barriers.

Explain how to identify impaired barriers and implement approved risk reduction measures for degraded or failed barriers.

Discuss and evaluate cumulative risk when multiple barriers are impaired or interdependent.

In the field, ask the front-line:

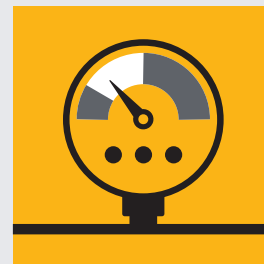
- What process safety barriers do you have at your site?
- Could this task impair any of those barriers? Are any barriers defeated during this procedure?
- If a barrier is impaired or defeated, how do you manage the risk?

Additional guidance:

- Energy Institute: Reflective Learning: [I own my barrier](#)
- Energy Institute: Reflective learning: [I keep my barrier strong](#)
- Step Change in Safety: [Own Your Barrier](#)
- US Chemical Safety Board: [Husky Energy Superior Refinery Explosion and Fire](#)
- US Chemical Safety Board: [Updated Texas City Animation](#)
- Safer Together Process Safety: [We All Have a Part to Play](#)

WE STAY WITHIN OPERATING LIMITS

- We discuss and use the approved operating limits for our location.
- We escalate where we cannot work within operating limits.
- We alert supervision if an alarm response action is unclear or the time to respond is inadequate.
- We obtain formal approval before changing operating limits.
- We confirm that potential for overpressure from temporary pressure sources has been addressed.



Equipment operated with process conditions (e.g., temperature, pressure, level, flowrate) outside of safe operating limits (high or low) can result in unstable and unpredictable operation and the potential for process safety incidents. It is therefore important to understand the operating limits of the plant and stay within that operating envelope. It is useful for teams to discuss how these limits are documented at their site and whether the limits are clear and available.

Overfilling and overpressure are the most common operating limit excursions that lead to process safety incidents. One common type of fatal process safety related incident occurs when a temporary source of high pressure (e.g., pump, compressor, nitrogen bottle, etc.) is connected to the process with inadequate overpressure protection. This can lead to catastrophic failure impacting those working in the area even if the release does not ignite. Teams are encouraged to think about occasions when temporary pressure sources are used at their facilities and confirm that strong process safety barriers are present. Consider that these are usually abnormal activities when regular barriers may not be in place.

IOGP Tier 1 process safety data also includes many examples of overfilling events. Confirming sufficient capacity for material transfers and monitoring the transfer whilst it progresses are important aspects of staying within the operating limits for level. Teams can discuss the potential for overfilling at their facility and whether monitoring transfers are realistic given other workload and distractions.

Some potential operating limit excursions are less obvious for example:

- Change in fluid composition – with the potential for corrosion and/ or erosion. This could be sudden (e.g., sand breakthrough from a well) or gradual (e.g., increasing water or hydrogen sulphide content of well fluids).
- Velocity changes due to lower operating pressure – leading to excessive vibration or erosion.

Teams can discuss other potential operating limit excursions relevant to their facility. Where there are concerns, advice should be sought from supervision and support groups. While there is always a desire to return to normal operation as soon as possible it is imperative to investigate and understand the causes of an excursion to prevent recurrence.

Tips for Leaders and Managers:

Establish and maintain documentation for safe operating limits of key process variables and make them visible to front-line workers.

Ensure that the potential consequences of excursions outside of safe operating limits are fully understood.

Systematically investigate excursions outside safe operating limits and ensure that abnormal conditions or situations can be returned to normal as soon as practical.

Regularly verify that personnel have the necessary skills and knowledge to stay within safe operating limits.

Demonstrate through decisions that cost, production, or schedule never override safety considerations.

In the field, ask the front-line:

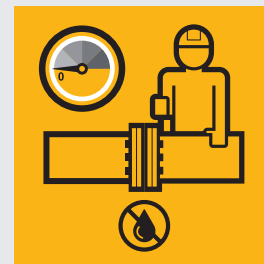
- What are the key operating limits associated with the task?
- Which alarms may be activated? If they activate what is the required response?

Additional guidance:

- IOGP Report 456 - *Process Safety – recommended practice on key performance indicators*
- Safer Together: [Safe Operating Envelope](#)
- Center for Chemical Process Safety Beacon (November 2007): [Cold Embrittlement and Thermal Stress](#)
- US Chemical Safety Board: [Texas City Refinery Explosion](#)

WE MAINTAIN SAFE ISOLATION

- We use isolation plans for the specific task, based on up-to-date information.
- We raise isolation concerns before the task starts and challenge when isolation plans cannot be executed.
- We check for residual pressure or process material before breaking containment.
- We monitor the integrity of isolations regularly and stop to reassess when change could affect an isolation's integrity.
- We confirm leak-tightness before, during, and after reinstating equipment.



It is important for every activity that requires breaking containment that an appropriate isolation plan for the specific activity is used and followed. Some process safety incidents have occurred when an isolation plan for a similar task has been used but did not address all the relevant hazards. Isolation plans should therefore match the task and be based on up-to-date process safety information (e.g., Piping and Instrumentation Diagrams (P&IDs)).

It is good practice to discuss isolation tasks and raise concerns before the task begins. This enables concerns to be considered and resolved safely.

If for some reason the isolations cannot be executed as planned, you should stop work and seek advice from your supervisor on how to proceed safely.

Concerns raised might include isolations that do not achieve positive isolation, quantities of drained materials that are more or less than expected, or indications of the presence of significant residual pressure or material.

After breaking containment, it is important to remain vigilant to potential signs that might indicate that the effectiveness of the isolations, venting, or draining arrangements have changed. This could include difficulties with proceeding with a task (e.g., due to trapped pressure or vacuum), more liquids arising, new smells, etc.

- Always make sure that you are either in sight of, or in control of the isolations you are using for your job. Before putting equipment back into service, it is important to verify that the system is mechanically complete using your local practices.
- Be vigilant about potential false pressure indications (e.g., line plugs, hydrates, etc.).
- Before introducing hydrocarbons, it is good practice to perform a gross leak test where possible.

Tips for Leaders and Managers:

Monitor how isolations are managed at your location to verify that the process is safe and effective, and implement improvements where issues are identified.

Review and verify that isolation plans are suitable for all tasks and test scenarios that could compromise the integrity of an isolation.

Respond promptly and follow up if isolation concerns are raised.

Periodically check the training and competence of those personnel performing isolations.

Regularly verify the status of secured or locked valves in the field and cross-check them against registers to maintain accuracy of valve positions.

In the field, ask the front-line:

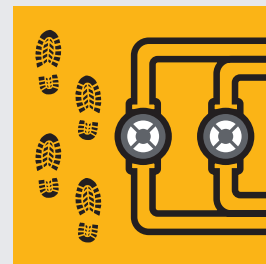
- What steps or actions will you undertake before breaking containment?
- How will you confirm leak-tightness before, during, and after reinstatement of equipment?

Additional guidance:

- UK Health and Safety Executive – [The safe isolation of plant and equipment](#)
- Step Change in Safety: [Hydrocarbon Release Prevention guidance](#)

WE WALK THE LINE

- We use up-to-date documentation (e.g., P&IDs) that accurately reflect installed systems and equipment.
- We physically confirm the system is ready for the intended activity (e.g., valve positions, line up of relief devices, etc.).
- We alert supervision to identified documentation and readiness issues before operation.



To avoid releases of hydrocarbons and other chemicals, we must confirm that process systems are ready for the next stage of operation. Many fatal incidents in IOGP process safety data have occurred when systems were put into service in a condition that was not as intended. This could be a valve inadvertently left in the wrong position, a drain or vent left unplugged, or a piping joint not fully tightened. It is important to check that systems are in good condition and correctly set-up each time we start them up or make a significant change in their use. This involves a physical, systematic inspection of the system in the field, sometimes in tandem with monitoring from the control room.

It is important that the process safety information (e.g., P&IDs) that we use to confirm readiness is accurate and up-to-date.

Physically checking systems is not always easy due to difficulties in access or visibility. Teams are encouraged to discuss the challenges of performing thorough 'walk-the-line' checks and alert supervision to readiness or documentation concerns.

Examples of when to walk-the-line include:

- Material transfers.
- Equipment changes.
- Handover from another work group.
- Manual change in operational state - equipment idled on another shift.
- Start-up after prolonged outage.
- Changing direction of flow.
- Return to operation after maintenance or turnaround.
- Changeover of equipment (e.g., relief valves, filters, pumps, compressors).
- Batch operations (e.g., pigging, bed regeneration).
- Temporary pipework in wells operations.

What should we do: Physically verify the system is ready for the intended operation, for example:

- P&IDs and other relevant information are consulted.
- Equipment and safeguards are installed correctly and functioning.
- Temporary equipment, hoses, etc., are suitable for use and in good condition.
- Plugs, drain valves, blinds, and spades are installed.
- Locked open/closed valves are in the correct position.
- Relief routes are not blocked.
- Unintended routes to atmosphere are isolated.
- Pressure is verified.
- If necessary, clear communication between the control room and the field is maintained (e.g., to confirm equipment tags before performing an operation).

Tips for Leaders and Managers:

Regularly confirm that process safety information is accurate and up-to-date.

Discuss 'walk-the-line' issues with front-line workers and follow up on concerns raised.

Participate in line walks to observe conditions firsthand.

Discuss findings from line walks and reinforce the important things to look out for during this activity.

In the field, ask the front-line:

- Why do process safety events often occur during startup activities?
- What are the key items you will physically check as part of a 'walk-the-line'?
- What would prevent you from restarting equipment after plant reinstatement?

Additional guidance:

- Center for Chemical Process Safety Beacon: [Walk the line \(animation\)](#), [Operational Readiness](#), [Operational Readiness Reviews](#)

WE CONTROL IGNITION SOURCES

- We identify, eliminate, or control the full range of potential ignition sources during task risk assessments and during job preparation and execution.
- We minimise and challenge ignition sources even in “non-hazardous” areas.
- We eliminate ignition sources during breaking containment and start-up and shutdown operations.



If a hydrocarbon release does occur, we can often reduce or eliminate the worst of the potential consequences by avoiding ignition. Where the fatal process safety incidents reviewed involved ignition, the number of fatalities per event was much higher, so minimizing the likelihood of ignition will make a big difference in avoiding fatalities. Some sources of ignition may be obvious, like hot work or nearby fired heaters, but others are less clear. The fatal incidents in the IOGP data show that ignition sources include hot work, static electricity, pyrophoric materials, ingestion into generators or other fired equipment, hot surfaces, lightning, defective electrical equipment and vehicles. Staff are therefore encouraged to understand the full range of potential ignition sources that are relevant to their facility and the measures in place to control them.

Use of hazardous area classification/zoning is widespread in the industry. But users should consider that the flammable gases from larger releases can extend far beyond the classified/zoned areas shown on the area classification drawings. It is important to recognize this, particularly if work occurs in areas that are outside the formally classified areas. The likelihood of getting a flammable cloud in these areas may be lower, but that does not mean it cannot happen. Risk assessments and risk control measures will need to take this into account.

In preparing for and conducting work activities, consider and act on the following:

- Ignition sources can include vehicles, open flames, grinding tools, pyrophoric materials, fixed and portable electrical equipment, hot surfaces, lightning, and static electricity.
- Reporting of defects in electrical equipment and other potential ignition sources, such as damaged insulation/cladding on hot surfaces, damaged EX equipment, open wires, damaged grounding aids, is important in controlling potential ignition.
- Vehicle entry into areas with the potential for flammable release to be avoided where practical. If that is not practical, vehicle entries should be minimized and controlled.
- Control of work risk assessments should evaluate the potential for flammable hazards even outside classified/zoned areas shown on the area classification drawing.

Tips for Leaders and Managers:

Regularly verify that personnel understand the full range of potential ignition sources and the requirements for ignition source control.

Verify temporary equipment in hazardous areas do not introduce ignition sources (hot surfaces, engine air intakes, non-ATEX rated circuits, static electricity potential, etc.).

Consider and discuss ignition sources for non-process flammable materials (e.g., methanol, injection chemicals, fuels, etc.).

In the field, ask the front-line:

- How far flammable gases could disperse if a release occurred?
- Which areas/equipment could the gas reach – what are the possible effects?
- Which equipment shuts down on gas detection, and which equipment should continue to operate during an emergency?

Additional guidance:

- Codes and Standards: IEC 60 079 series, ANSI/API RP 505, EI Model Code of Safe Practice 15 - *Area classification for installations handling flammable fluids*
- Center for Chemical Process Safety Beacon (August 2016): [A little static can cause a big fire!](#)
- Center for Chemical Process Safety Beacon (August 2014): [Ignition Sources](#)
- US Chemical Safety Board: [Fire at Evergreen Packaging](#)

WE RECOGNISE CHANGE

- We look for and speak up about change.
- We discuss changes and involve others to identify the need for management of change (MOC).
- We review the MOC process for guidance on what triggers an MOC.
- We discuss and seek advice on change that occurs gradually over time.



Management of Change (MOC) failure has long been recognized in the oil and gas industry as a cause of process safety incidents. It is normal for companies to have a MOC process in place, but despite this, the IOGP fatality data indicates that MOC continues to be linked to many fatal accidents. This is typically when a change was made without passing it through the MOC process, meaning that it did not get thoroughly reviewed and risk assessed. This can be because it was not identified that the change needed an MOC, so it is important that teams understand and are sensitive to what changes need to pass through the MOC process. This enables them to look out for change and speak up if they identify a change that they think should be managed by the MOC process.

Whilst it is expected that companies will have an MOC process in place already, this PSF focuses on recognizing change that needs to be subject to MOC. Sometimes changes that might appear quite small can have a large impact on process safety, either by introducing new hazards or degrading existing barriers. It is good for teams to discuss changes at their location and to involve others to help identify whether formal MOC is necessary.

Change, which can be permanent, temporary or emergency, is not limited to hardware modifications and can typically include:

- Operating or maintenance procedure revisions
- Changes to operating limits
- Changes to process operating conditions (e.g., temperature, pressure, flowrate, composition, pH/acidity)
- Equipment changes (non like-for-like replacement)
- Change of chemicals or materials

Teams can review their local MOC process for guidance on what triggers their MOC process and discuss examples at their location. It is recognized that there may be pressure to “get work done” but experience has shown that change should be systematically managed in all situations to avoid unwanted incidents.

It is also important to be alert to change that occurs slowly, perhaps over many years. This “creeping change” (e.g., gradual change in process fluid composition or gradual reduction in a team’s experience level) also needs evaluation. Making time to discuss creeping change, or perhaps the impact of multiple small changes on the same system, can be useful in avoiding incidents. Teams can seek advice or alert supervision if they have concerns.

Tips for Leaders and Managers:

Verify that systematic identification and MOC processes are in place and working effectively.

Verify that workforce personnel and management are trained and competent to recognize and respond to changes.

In the field, ask the front-line:

- Has anything changed since you last performed or were trained on this task? If so, how might that affect the way you do it?
- Have there been any recent changes to the site that could impact your task or daily activities?
- How are these changes being managed or accounted for?

Additional guidance:

- Energy Institute, CCPS, OSHA, Process Safety Management Systems
- Center for Chemical Process Safety Beacon (July 2017): [Management of Change](#)
- Center for Chemical Process Safety Beacon (September 2016): [Can you recognize a change?](#)
- Center for Chemical Process Safety Beacon (October 2012): [Manage Temporary Changes!](#)
- US Chemical Safety Board: [Fire from ice](#)
- US Chemical Safety Board: [Blocked in](#)

WE STOP IF THE UNEXPECTED OCCURS

- We discuss the work plan and what signals would tell us it is proceeding as expected.
- We pause and ask questions when signals and conditions are not as expected.
- We stop and alert supervision if the activity is not proceeding as expected.



We normally plan our activities carefully, thinking about how we expect the activity to proceed and what could potentially go wrong. However, in real life there are times when things do not occur as planned or expected.

Often process safety incidents have occurred when an activity has deviated from the expected path, but those involved have continued anyway or informally adapted the plan on-the-run. While this may be well-intended to get the job done, it has the potential to lead down a path that has not been fully evaluated and risk assessed.

This process safety fundamental highlights the importance of recognizing when things are not progressing as expected and being disciplined about stopping or pausing to understand what has changed and why things are not as planned. Often this may need a change to the plan or an update of the risk assessment, or it may prompt consultation with others to seek specialist input. We should be ready to speak up and challenge unexpected situations and engage with supervision to discuss our concerns. Pausing or stopping a task may feel inconvenient at the time but it may avoid a process safety event, and by doing so, prevent fatalities or injuries.

Tips for Leaders and Managers:

Empower everyone, to stop and seek guidance if they are unsure, or if something doesn't feel right, or if an activity is not proceeding as expected.

Discuss and explain valid reasons for stopping work, such as:

- Missing or unclear information.
- Poor communication.
- Human factors: fatigue, ergonomics, unsuitable tools, time pressure.

Positively recognize people who pause to reassess when things aren't going as expected.

In the field, ask the front-line:

- What checks or actions do you undertake during a task to verify it's proceeding as expected? What signals are you looking for to verify this?
- What signals or conditions suggest a task is not proceeding as expected?
- What challenges do you face when deciding to stop work?

Additional guidance:

- Center for Chemical Process Safety Beacon (July 2014): [Persistence – Good or Bad?](#)
- Step Change In Safety: [MAH Understanding Vlog: Situation Awareness](#)
- US Chemical Safety Board: [Investigations of Incidents during Start-ups and Shutdowns](#)

WE WATCH FOR WEAK SIGNALS

- We proactively look for indicators or signals that suggest future problems.
- We speak up about potential issues even if we are not sure they are important.
- We persistently explore the causes of changing indicators or unusual situations.



One characteristic of most major process safety incidents is that before they occurred, there were noticeable indications that a problem was developing. Examples of weak signals include unusual vibration, ice unexpectedly forming on the outside of a pipe, weeps and seeps, passing valves, reoccurring alarms, or abnormal field readings. It is good practice for teams to discuss examples of weak signals that they have seen in their facility.

The key is to identify these “weak signals” and respond to them before something more serious occurs. Front-line workers are often considered best placed to pick up on weak signals as they become very familiar with what is normal and what is not. Personnel are therefore encouraged to remain alert to these signals, even if they seem unimportant, so that they can be discussed and evaluated.

Operator rounds, visual inspections, or just going about our routine work at site, are times when we can use our eyes and ears to remain alert to anything that seems out of place or different.

When weak signals are identified and reported it can be tempting to dismiss them too readily. Weak signals can be a vital opportunity to act early to avoid an accident, so it is important that they are persistently explored to understand the cause.

It is important to remain constantly aware of the potential for process safety incidents. Some companies describe this as maintaining a state of “chronic unease”. Always being aware of our vulnerability to a process safety event can help us to avoid having one.

Tips for Leaders and Managers:

Stay alert to weak signals – small signs that something might not be right and respond proactively when they are raised. Encourage open communication of all process safety concerns to management.

Recognize those who identify process safety weak signals.

Review procedures after completing a task and use feedback to reduce the risk of future errors. Consider:

- Adding steps that require a pause or confirmation before moving to the next stage.
- Ensuring the previous step is verified as complete before proceeding.

In the field, ask the front-line:

- What are weak signals, and why do they matter?
- What action would you take if you notice a weak signal?
- What process safety performance indicators do we monitor? Discuss to build understanding.
- Do you have any examples where acting on weak signals prevented an undesirable condition or incident?

Additional guidance:

- Energy Institute Reflective Learning Tools: [Chronic Unease](#)
- International Association of Oil and Gas Producers: [Weak signals video](#)

5. Definitions

Key steps in operation or maintenance procedure – A procedure step, series of steps, or action that, if performed improperly, will cause irreversible harm to plant, equipment, or people or will significantly impact plant operation. It is also known as “critical steps”.

Safe isolation – The separation of plant and equipment from every source of energy (pressure, electrical, and mechanical) in such a way that the separation is secure. Non-proven isolation methods (i.e., without provision to confirm effectiveness of the isolation) should not be considered a safe isolation when dealing with hazardous conditions (e.g., hazardous substances, pressure, temperature, etc.).

Hazardous material – Materials that can cause harm, for example by igniting or being toxic. API 754 and IOGP Report 456 - *Process safety - recommended practice on key performance indicators* refer to the United Nations Recommendations on the Transport of Dangerous Goods.

Like-for-like equipment change – changes that are a “replacement in kind” which means to replace with the same design, same rating, same materials of construction, etc.

Safe Operating Limit (SOL) – Limits established for process parameters, such as temperature, pressure, level, flow, or concentration, based on a combination of equipment design limits and the dynamics of the process. For further reading refer to IOGP Report 456.

Walk the line – Walk the line is a practice sharing programme that provides a variety of tools to help ensure that operators know where energy will flow between two points in a process. Typical activities for walk the line to be applied:

- Return to operation after maintenance or turnaround.
- Changeover of equipment (e.g., relief valves, filters, pumps, compressors).
- Transfer of fluids (e.g., loading operations).
- Sequential or batch operations (e.g., pigging, bed regeneration).
- Temporary pipework.

Hardware Barrier – Primary containment, process equipment and engineered systems designed and managed to prevent Loss of Primary Containment (LOPC) and other types of asset integrity or process safety events, and mitigate any potential consequences of such events. These are checked and maintained by people (in critical activity/tasks). For further reading refer to IOGP Report 544 - *Standardization of barrier definitions*.

Human Barriers – Barriers that rely on the actions of people capable of carrying out activities designed to prevent LOPC and other types of asset integrity or process safety events and mitigate any potential consequences of such events. For further reading refer to IOGP Report 544.

Weak signals – Weak signals are ‘tell-tale’ signs, which are not considered normal operating conditions of a given plant (unusual noise or vibration; pressure or level readings that are not as expected, etc.). There are various methods to identify and report “weak signals” such as operator rounds, near miss programmes, etc. A constant status of alertness to weak signals, such as asking the right questions and picking up on signals of potential failure when on-site is a key behaviour for safety leaders at every level in the organization to help reduce the risk of incidents.

Risk normalization – The tendency to underestimate a risk that has become familiar and is associated with tasks that we undertake regularly without incident. If someone deviates from safe practices on one occasion, it increases the likelihood this behaviour will be repeated, until this unsafe behaviour becomes routine and accepted. This gradual justification and acceptance of unsafe behaviour can happen to any individual or organization and vigilance is required to ensure that established safe practice is maintained.

Hazardous Area Classification (HAC) – Any place an ignitable concentration of flammable gases or vapours may occur in quantities that would require special precautions to protect the safety of workers and plant. In this context, ‘special precautions’ are best taken as relating to the construction, installation, and use of apparatus. It should be noted there is limitation with HAC, whereby migration of flammable material (gas dispersion) can occur beyond limits of HAC.

Abnormal situation – An abnormal situation is a disturbance or series of disturbances in a process that causes plant operations to deviate from their normal operating state (for further reading on normal operating limits refer to IOGP Report 456 – *Process safety – recommended practice on key performance indicators*). The disturbances may be minimal or catastrophic, and may cause production losses or, in serious cases, endanger human life.

Up-to-date P&IDs – Updated set of drawings Piping and Instrumentation Diagram (P&ID) reflecting the current and real plant status. Usually submitted by a contractor upon completion of a construction project and used in the operations phase. Depending on the plant status these can also be known as Red-Line Mark-Up or As-Built P&IDs.

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Designed to support those working in front-line operations, maintenance, and well related activities, the IOGP Process Safety Fundamentals are informed by data and designed to draw attention to situations that are most likely to lead to process safety event fatalities. They are not intended to exhaustively address all process safety risks and hazards in the oil and gas industry, but to be deployed in addition to a company's underlying systems for process safety management.

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